

Show all steps or explain how you found your answer. Use digital inking for each question involving mathematical symbols. If the answer is a sentence with no mathematical symbols (like question 3), feel free to type your response to that question. However, do not mix digital inking with typing.

1. Write the following statements as a combination of p and q using propositional logic where

p : I will do all of my homework

q : I will get an A in Discrete Math

- a. I will get an A in Discrete Math **only if** I do all of my homework (5 pts)

$$q \rightarrow p$$

- b. **Either** I will not get an A in Discrete Math **or** I will **not** do my homework (5 pts)

$$\neg q \vee \neg p$$

- c. **Either if** I **don't** do all of my homework, **then I won't** get an A in Discrete Math, **or** I will do all of my homework **and** get an A in Discrete Math. (5 pts)

$$(\neg p \rightarrow \neg q) \vee (p \wedge q)$$

2. Show that $\neg(\neg p \vee q) \equiv (\neg q \wedge p)$

- a. using a truth table (5 pts)

p	q	$\neg p$	$\neg p \vee q$	$\neg(\neg p \vee q)$	$\neg q$	$(\neg q \wedge p)$
T	T	F	T	F	F	F
T	F	F	F	T	T	T
F	T	T	T	F	F	F
F	F	T	T	F	T	F

- b. using logical equivalences (5 pts)

Step 1: $\neg(\neg p \vee q)$

Reason: Premise

Step 2: $\neg(\neg p) \wedge \neg q$

Reason: De Morgan's Law on (1)

Step 3: $p \wedge \neg q$

Reason: Double Negation Law on (2)

Step 4: $\neg q \wedge p$

Reason: Commutative Law on (3)

Thus, $\neg(\neg p \vee q) \equiv \neg q \wedge p$

3. What is the converse, inverse and contrapositive of the statement "If I get a tax refund, then I will go on vacation"?

a. Assign each proposition and write the original statement using those propositions. (5 pts)

Let p be "I get a tax refund."

Let q be "I will go on vacation."

Original Statement using those propositions:

$$p \rightarrow q$$

b. Converse (5 pts)

$$q \rightarrow p$$

"If I go on vacation, then I got a tax refund."

c. Inverse (5 pts)

$$\neg p \rightarrow \neg q$$

"If I do not get a tax refund, then I will not go on vacation."

d. Contrapositive (5 pts)

$$\neg q \rightarrow \neg p$$

"If I do not go on vacation, then I will not get a tax refund."

4. Suppose that $Q(x)$ is the propositional function " $5x + 2 = x$." What are the truth values for $\forall x Q(x)$ and $\exists x Q(x)$ for the domain of all real numbers? Explain! (5 pts)

Given: $Q(x): 5x + 2 = x$ for the domain of all real numbers.

Solve the equation: $5x + 2 = x \rightarrow 4x + 2 = 0 \rightarrow 4x = -2 \rightarrow x = -\frac{1}{2}$

So, the equation is only true when $x = -\frac{1}{2}$.

Conclusion: 1.) $\forall x Q(x)$ means "For all real numbers x , $5x + 2 = x$." This is false because the equation does not hold for all real numbers, only $x = -\frac{1}{2}$.

2.) $\exists x Q(x)$ means "There exists a real number x such that $5x + 2 = x$." This is true because we found a solution of $x = -\frac{1}{2}$.

5. Prove or disprove each of the following statements. Use any method. (10 pts each)

a. The sum of two odd integers is always even.

Given: The sum of two odd integers. WTP: The sum of two odd integers is always even.

Proof: Let m and n be odd integers. By definition of an odd integer, there exists integers a and b such that, $m = 2a + 1$ and $n = 2b + 1$. Now consider their sum:

$$m + n = (2a + 1) + (2b + 1) = 2a + 2b + 2 = 2(a + b + 1).$$

Since a and b are integers, then $(a + b + 1)$ is an integer. Let $c = a + b + 1$.

$$\text{Then, } m + n = 2c.$$

By definition, an integer in the form $2c$ is even.

Conclusion: $\therefore$ The sum of two odd integers is always even. ■

b. The product of an even integer and an even integer is always even.

Given: The product of two even integers. WTP: The product of two even integers is always even.

Proof: Let m and n be two even integers. By definition of an even integer, there

exists integers a and b such that, $m = 2a$ and $n = 2b$.

Then, $m \cdot n = (2a)(2b) = 4ab = 2(2ab)$.

Since a and b are integers, then $2ab$ is an integer. Let $c = 2ab$.

Then, $m \cdot n = 2c$.

By definition of an integer, an integer in form $2c$ is even.

Conclusion: $\therefore$ The product of two even integers is always even. ■

c. 10 is not the square of a positive integer (Be sure your proof covers ALL integer values of x . To do this, use proof by cases where Case 1 is $1 \leq x \leq 3$ and Case 2 is $x \geq 4$)

Given: x is a positive integer. WTP: There is no $x \in \mathbb{Z}^+$ such that $x^2 = 10$.

Proof: Case 1: $1 \leq x \leq 3$. Since x is a positive integer in this range, the only possibilities are $x = 1, 2, 3$.

If $x = 1$, then $x^2 = 1 \neq 10$.

If $x = 2$, then $x^2 = 4 \neq 10$.

If $x = 3$, then $x^2 = 9 \neq 10$.

In each case $x^2 \leq 9$, which $x^2 \neq 10$ for all x , $1 \leq x \leq 3$.

Case 2: $x \geq 4$. Because x is an integer and $x \geq 4$, then $x^2 \geq 4^2 = 16$. Thus,

$x^2 \geq 16 > 10$. So, $x^2 \geq 16$ and $x^2 \neq 10$ for all $x \geq 4$.

All $\mathbb{Z}^+$, x , have been covered with both cases. In case 1, $x^2 \leq 9$ and $x^2 \neq 10$ and in case 2, $x^2 \geq 16$ and $x^2 \neq 10$.

Conclusion: $\therefore$ 10 is not the square of a positive integer. ■

d. There are no solutions in positive integers to the equation $x^4 + y^4 = 1000$.

Given: x and y are $\mathbb{Z}^+$. WTP: There are no $\mathbb{Z}^+$ solutions to $x^4 + y^4 = 1000$.

Proof: First, note all the small 4th powers: $1^4 = 1$, $2^4 = 16$, $3^4 = 81$, $4^4 = 256$, $5^4 = 625$, $6^4 = 1296$.

Since $6^4 = 1296 > 1000$, neither x nor y can be ≥ 6 . Thus, $x, y \leq 5$.

Case 1, $\{x, y\} = 5$: WLOG, let $x = 5$. Then, $y^4 = 1000 - x^4 = 1000 - 625 = 375$. But there is no integer y such that $y^4 = 375$, since $4^4 = 256 < 375 < 625 = 5^4$. Thus, there is no solution in this case.

Case 2, $\{x, y\} \leq 4$: Then, $x^4 + y^4 \leq 4^4 + 4^4 = 256 + 256 = 512 < 1000$, so $x^4 + y^4$ cannot equal 1000 in this case either. Both cases exhaust all $\mathbb{Z}^+$ x, y . Neither case holds that $x^4 + y^4 = 1000$.

Conclusion: $\therefore$ There are no solutions for $\mathbb{Z}^+$ such that $x^4 + y^4 = 1000$. ■

6. Use rules of inference to show that the premises $p \vee q, q \rightarrow r, r \rightarrow s, \neg p$ lead to a conclusion of s .

<u>Step</u>	<u>Reason</u>
1. $p \vee q$	Premise
2. $q \rightarrow r$	Premise
3. $r \rightarrow s$	Premise
4. $\neg p$	Premise
5. q	Disjunctive Syllogism from (1) and (4).
6. r	Modus Ponens from (2) and (5).
7. s	Modus Ponens from (3) and (6).

Bonus (5 pts) $\therefore s$ follows from the premises $p \vee q, q \rightarrow r, r \rightarrow s$, and $\neg p$. ■

Give the output, Q , of the following logic gate, including all intermediate steps.

